

S-SPPO: Semantic-Calibrated Self-Play Preference Optimization

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Abstract

Aligning Large Language Models (LLMs) with human preferences is often formulated via Direct Preference Optimization (DPO). However, the standard Bradley-Terry instantiation of DPO is limited in modeling common departures from transitivity in human preferences. To address this, recent work has introduced Self-Play Preference Optimization (SPPO), which iteratively refines the policy by training on self-generated win-lose pairs. Our investigation, however, reveals a critical instability in SPPO: the optimization is prone to *policy degeneration* when the preference oracle assigns overly confident wins to semantically indistinguishable responses. To mitigate this, we propose S-SPPO, a dual-space semantic calibration framework comprising: i) *Supervision Calibration* via semantic gating, which anneals win rate targets toward the maximum-entropy baseline as semantic overlap increases; and ii) *Representation Calibration* via latent repulsion to enforce geometric diversity to prevent manifold collapse and maintain latent diversity between chosen and rejected samples. Theoretically, we show that the calibration preserves the constant-sum game structure, facilitating convergence to a Nash Equilibrium. Empirically, S-SPPO avoids the performance degradation seen in prior methods, achieving **52.19% win rate** and **47.46% length-controlled win rate** on AlpacaEval 2.0 with Llama-3-8B, without using additional human-annotated preferences during training. The code will be available at <https://github.com/xiwencl/s-sppe>.

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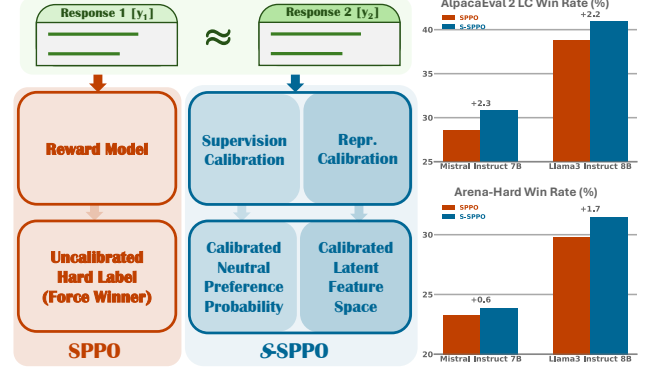


Figure 1. Comparison between S-SPPO and SPPO. When responses (y_1, y_2) are semantically similar, SPPO imposes uncalibrated hard preferences that can be arbitrary, whereas S-SPPO employs *Supervision* and *Representation* calibration to produce calibrated preference probabilities (left), resulting in more stable training and consistent empirical improvements (right).

1. Introduction

The alignment of Large Language Models (LLMs) with human values has become a cornerstone of modern AI development. (Ouyang et al., 2022; Bai et al., 2022; Touvron et al., 2023; Sun et al., 2024; Chen et al., 2025) The standard paradigm, Reinforcement Learning from Human Feedback (RLHF, Christiano et al., 2017; Ouyang et al., 2022), typically involves training a reward model to proxy human preferences, followed by optimizing the policy via Proximal Policy Optimization (PPO, Schulman et al., 2017). While effective, this pipeline is often criticized for its complexity, training instability, and sensitivity to hyperparameter tuning (Engstrom et al., 2020; Azar et al., 2024; Casper et al., 2023; Rafailov et al., 2023).

To mitigate these challenges, Direct Preference Optimization (DPO, Rafailov et al., 2023) emerged as a compelling alternative. By reparameterizing the optimal policy in closed form, DPO bypasses the need for an explicit reward model, treating preferences as an implicit reward signal to be optimized directly. However, the standard DPO formulation relies on the *Bradley-Terry* (BT) model (Bradley & Terry, 1952), which assumes that preferences satisfy strict transi-

tivity and can be fully captured by a scalar reward function. This assumption, however, often fails to account for the *intransitivity* and irrationality inherent in general human preferences (Wu et al., 2025).

To bridge this gap, Self-Play Preference Optimization (SPPO, Wu et al., 2025) reframes the alignment problem through the lens of game theory. Instead of maximizing a static reward, SPPO formulates preference optimization as a two-player constant-sum game between the policy and the environment. By approximating the Nash Equilibrium, SPPO theoretically enables the model to handle non-transitive preferences robustly, without being constrained by the limitations of the Bradley-Terry model.

However, SPPO implicitly assumes a well-calibrated preference oracle. Specifically, the algorithm treats estimated win rates as precise regression targets for policy optimization. This assumption breaks down in practice. We observe that practical reward models are often highly overconfident, assigning extreme win probabilities (approaching 0 or 1) even when candidate responses are semantically indistinguishable. *This introduces a critical misalignment in the optimization landscape.* Since SPPO forces the model to fit these overconfident targets, the optimizer effectively tries to minimize the residual between the implicit reward (*i.e.*, log-likelihood ratio) and noise in the preference predictions. Consequently, the training signal becomes dominated by these *false positives* on semantically indistinguishable pairs, aggressively pushing the model to distinguish between highly similar outputs. This artificial scaling of gradient variance destabilizes the self-play trajectory, preventing convergence to a meaningful Nash Equilibrium.

To bridge this gap, we propose Semantic-Calibrated Self-Play Preference Optimization (S-SPPO), a dual-space semantic calibration framework designed to stabilize self-play dynamics. Conceptually, we reframe preference optimization as a *Dual-Space Min-Max game*. We identify that stable alignment requires satisfying adversarial constraints: *minimizing* preference divergence in the label space to prevent overfitting noise, while simultaneously *maximizing* representation distance in the latent space to prevent manifold collapse (see Fig. 1(a)). This leads to consistent performance improvement in various benchmarks (Fig. 1(b)).

To summarize, our main contributions are as follows:

- We uncover a critical instability in self-play preference optimization (SPPO), revealing that the optimization is prone to learning instability and performance degradation when the preference oracle assigns overly confident wins to semantically indistinguishable responses.
- We propose S-SPPO, which introduces two novel calibration mechanisms: *i) Supervision Calibration* via semantic gating to anneal win rate targets based on

semantic overlap; and *ii) Representation Calibration*, which applies a repulsive force in the latent space to prevent mode collapse.

- We provide a theoretical analysis proving that our semantic calibration preserves the constant-sum game structure, thereby facilitating convergence to a Nash Equilibrium comparable to the original SPPO formulation.
- Empirically, S-SPPO significantly outperforms existing baselines. On Llama-3-8B, it achieves a 52.19% win rate and a 47.46% length-controlled win rate on AlpacaEval 2.0 without relying on additional human-annotated preference data or supervision from stronger models (e.g., GPT-4) during training, effectively eliminating the performance degradation observed in standard self-play methods.

2. Background and Limitations of SPPO

2.1. Preference Optimization Landscape

RLHF (Ouyang et al., 2022) has emerged as a promising paradigm for aligning LLMs with human preference. Traditional RLHF approaches (*e.g.*, Schulman et al., 2017) typically follow a two-stage process. Given a prompt $\mathbf{x} \sim \mathcal{X}$, where $\mathcal{X}$ denotes the prompt distribution, an LLM policy $\pi(\cdot | \mathbf{x})$ generates a response $\mathbf{y} \in \mathcal{Y}$. The model aims to update the policy π to better match the human-preferred response distribution. The optimization objective is usually formulated as maximizing the expected reward subject to a Kullback–Leibler (KL) divergence constraint:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{X}} [r_\phi(\mathbf{y}; \mathbf{x})] - \beta \mathbb{E}_{\mathbf{x} \sim \mathcal{X}} [\text{KL}(\pi(\cdot | \mathbf{x}) \| \pi_{\text{ref}}(\cdot | \mathbf{x}))],$$

where π_{ref} is the reference policy (typically the Supervised Fine-Tuned model) and β controls the deviation penalty. While effective, traditional RLHF methods relying on explicit reward modeling often overlook the inherent semantic ambiguity in language.

Direct Preference Optimization. DPO (Rafailov et al., 2023) bypasses explicit reward modeling by treating preferences as an implicit reward signal, substantially simplifying RLHF for preference optimization. As is standard, we assume access to an offline preference dataset $\mathcal{D} = \{(\mathbf{x}, \mathbf{y}_1, \mathbf{y}_2)\}$ comprising preferred response $\mathbf{y}_1$ and dispreferred response $\mathbf{y}_2$ for prompt $\mathbf{x}$. DPO maximizes the log-likelihood $\mathbb{P}(\mathbf{y}_1 \succ \mathbf{y}_2 | \mathbf{x})$ under the Bradley-Terry (BT) preference model (Bradley & Terry, 1952). However, the off-policy DPO is known to fall short in capturing the intransitivity and irrationality in human preferences (Wu et al., 2025).

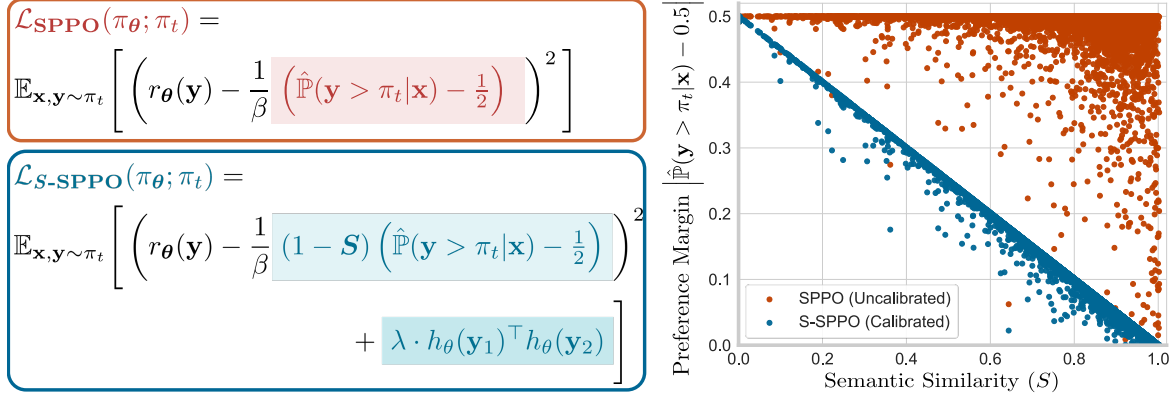


Figure 2. **S-SPPO** and **SPPO** mainly differ in their preference optimization objectives, as indicated in the shaded boxes (left). The preference margin $|\hat{\mathbb{P}}(\mathbf{y} > \pi_t | \mathbf{x}) - 0.5|$ in **SPPO** is uncalibrated, whereas **S-SPPO** calibrates the preference margin by the semantic similarity (S) between win-lose pairs (right). See Appendix C.1 for more details.

Self-Play Preference Optimization. SPPO (Wu et al., 2025) mitigates DPO’s limitations by optimizing preference probabilities directly in a self-play setting. It recasts preference optimization as a two-player constant-sum game between a policy π and an opponent π' , defining the expected win rate as

$$\mathbb{P}(\pi \succ \pi' | \mathbf{x}) \triangleq \mathbb{E}_{\substack{\mathbf{y} \sim \pi(\cdot | \mathbf{x}) \\ \mathbf{y}' \sim \pi'(\cdot | \mathbf{x})}} [\mathbb{P}(\mathbf{y} \succ \mathbf{y}' | \mathbf{x})].$$

The policy targets the *von Neumann winner* π^* , which is the Min–Max optimal strategy maximizing the win rate against any adversary:

$$(\pi^*, \pi^*) = \arg \max_{\pi} \min_{\pi'} \mathbb{E}_{\mathbf{x} \sim \mathcal{X}} [\mathbb{P}(\pi \succ \pi' | \mathbf{x})].$$

To approximate π^* , SPPO iteratively updates the policy. At iteration t , an online preference dataset $\mathcal{D}_t$ is constructed by generating K responses and annotating win-lose pairs using an off-the-shelf reward model (e.g., PairRM, Jiang et al., 2023b). Rather than maximizing a contrastive margin, SPPO treats the empirical win rate as a regression target. It updates the policy by minimizing the L_2 distance between the implicit reward $r_\theta(\mathbf{y})$ (log-likelihood ratio) and the shifted win rate:

$$\mathcal{L}_{\text{SPPO}}(\pi_\theta; \pi_t) = \mathbb{E}_{\mathcal{D}_t} \left[\left(\underbrace{\log \frac{\pi_\theta(\mathbf{y} | \mathbf{x})}{\pi_t(\mathbf{y} | \mathbf{x})}}_{r_\theta(\mathbf{y})} - \frac{1}{\beta} \left(\hat{\mathbb{P}}(\mathbf{y} \succ \pi_t | \mathbf{x}) - \frac{1}{2} \right) \right)^2 \right]. \quad (2.1)$$

This objective effectively regresses the policy’s log-ratio towards the preference probability, pushing the likelihood of winning responses away from losing ones to approximate the Nash Equilibrium dynamics.

2.2. Limitations of SPPO

The success of SPPO hinges on the reliability of the preference supervision signals $\mathcal{D}_t$ used for policy update at every iteration. Ideally, the preference oracle should reflect the underlying *epistemic uncertainty*: when two candidate responses are semantically indistinguishable, the estimated win rate should naturally converge to the neutral point 0.5. In other words, the *preference margin* $m(\mathbf{y}) = |\hat{\mathbb{P}}(\mathbf{y} \succ \pi_t | \mathbf{x}) - 0.5|$ should approach zero. To reveal this, we empirically investigate the relationship between the semantic similarity between win-lose pairs and the preference margin. However, we uncover a critical misalignment between the semantic similarity (S) of win-lose pairs and the reward model’s induced preferences (Fig. 2(right)). In particular, the reward model (e.g., PairRM) is poorly calibrated: standard SPPO continues to produce large preference margins even when the two responses are semantically indistinguishable (e.g., $S > 0.9$). As a result, the preference signal of SPPO is not aligned with semantic similarity, as highly similar responses can still be assigned extreme win probabilities, causing the optimization to treat negligible differences as substantive and to apply overly strong policy updates. This behavior indicates that discrete preference supervision fails to respect the inherently continuous structure of semantic similarity, effectively imputing substantial quality gaps where none exist. Consequently, the SPPO training signal can become decoupled from semantic consistency, degrading learning dynamics: to match high-confidence targets on ambiguous pairs, the policy may overfit to spurious cues or reward-model noise. Moreover, magnifying these high-magnitude errors can increase gradient variance, destabilize optimization, and hinder convergence toward a semantically consistent Nash Equilibrium. The gradient

analysis of SPPO reveals this fact

$$\nabla_{\theta} \mathcal{L}_{\text{SPPO}}(\pi_{\theta}; \pi_t) = 2 \underbrace{\left(r_{\theta}(\mathbf{y}) - \frac{m(\mathbf{y})}{\beta} \right)}_{w(\mathbf{y})} \cdot \nabla_{\theta} \log \pi_{\theta}(\mathbf{y}|\mathbf{x}), \quad (2.2)$$

where the term $\nabla_{\theta} \log \pi_{\theta}(\mathbf{y}|\mathbf{x})$ increases the log-probability of the chosen response, and the magnitude of $w(\mathbf{y})$ determines how hard SPPO pushes towards that direction. This analysis highlights the potential for performance degradation, which is consistent with the downward trends observed in Table 3 and Table 4.

3. Method: Dual-Space Semantic Calibration

Building on the analysis above, we propose S-SPPO, a framework that casts preference optimization as a *Dual-Space Min-Max game*. The framework enforces adversarial constraints at two complementary levels: *i)* Minimization: in the label space, Supervision Calibration minimizes preference divergence on ambiguous pairs, mitigating supervision bias; and *ii)* Maximization: in the latent space, Representation Calibration enforces geometric diversity between responses, preventing manifold collapse. Together, this push-pull dynamic stabilizes the policy under noisy supervision while preserving the expressiveness required for better alignment.

3.1. Supervision Calibration via Semantic Gating

The raw empirical win rate $\hat{\mathbb{P}}$, computed from reward-model scores, is often poorly calibrated with respect to semantic utility. To mitigate this, we introduce a semantic gating mechanism. Let $\phi(\cdot)$ denote a model-agnostic semantic encoder (Wang et al., 2020; Günther et al., 2023; Nussbaum et al., 2024), and define the *rectified semantic similarity* between responses $\mathbf{y}_i$ and $\mathbf{y}_j$ as follows:

$$S_{ij} = \max \left(0, \frac{\phi(\mathbf{y}_i)^{\top} \phi(\mathbf{y}_j)}{\|\phi(\mathbf{y}_i)\| \|\phi(\mathbf{y}_j)\|} \right).$$

We then obtain a calibrated target probability $\hat{\mathbb{P}}_c$ by annealing $\hat{\mathbb{P}}$ toward the maximum-entropy baseline (i.e., 0.5)¹, in proportion to each candidate’s average semantic overlap with the other responses:

$$\begin{aligned} \hat{\mathbb{P}}_c(\mathbf{y}_i \succ \pi_t|\mathbf{x}) &= \frac{1}{2} + \left(1 - \frac{1}{K-1} \sum_{j \neq i} S_{ij} \right) \\ &\quad \times \left(\hat{\mathbb{P}}(\mathbf{y}_i \succ \pi_t|\mathbf{x}) - \frac{1}{2} \right). \end{aligned} \quad (3.1)$$

¹Following Wu et al. (2025), we set the baseline target to 0.5. While originally derived as a normalization under random preferences, in our semantic calibration setting it naturally corresponds to the maximum-entropy case with no preference between semantically indistinguishable responses.

This formulation ensures the supervision signal is calibrated to semantic similarity: gradients are masked ($\hat{\mathbb{P}}_c \rightarrow 0.5$) when a response is semantically indistinguishable from its competitors, preventing the policy from fitting label noise.

Practical Implementation. While Eq. (3.1) provides a general formulation considering the semantic density over K samples, computing the full similarity matrix can be computationally expensive for large K . In our practical implementation (Algorithm 1), we align with the standard SPPO training paradigm, which samples a winner-loser pair $(\mathbf{y}_w, \mathbf{y}_l)$ for optimization. Accordingly, we approximate the average semantic overlap using the specific similarity between the selected pair, i.e., $S \approx S_{wl}$. This serves as an efficient Monte Carlo estimate of the semantic ambiguity, ensuring that the calibrated target $\hat{\mathbb{P}}_c$ effectively penalizes overconfidence when the chosen pair is semantically ambiguous.

Connection to Tilted-ERM. Our approach relates to Tilted-ERM (Li et al., 2020) in suppressing outliers. However, unlike TERM which performs blind re-weighting based on loss magnitude, S-SPPO utilizes semantic similarity as an informative prior. This selectively dampens signals for ambiguous pairs while retaining supervision for semantically distinct yet challenging instances.

3.2. Representation Calibration via Latent Repulsion

While Supervision Calibration mitigates label noise, frequent annealing of target probabilities to 0.5 introduces a regularization trade-off: the policy may lose discriminative power. This creates a risk of *manifold collapse*, where distinct generation trajectories map to near-identical latent representations due to gradient starvation. We propose Representation Calibration to enforce geometric diversity. Even for semantically ambiguous pairs, the model must maintain distinct internal states for different trajectories. Let $h_{\theta}(\mathbf{y})$ be the normalized final hidden state of an LLM. We define the calibration term on sampled pairs $(\mathbf{y}_1, \mathbf{y}_2)$ as:

$$\mathcal{L}_{\text{Rep}}(\theta) = \mathbb{E}_{(\mathbf{x}, \mathbf{y}_1, \mathbf{y}_2) \sim \mathcal{D}_t} \left[h_{\theta}(\mathbf{y}_1)^{\top} h_{\theta}(\mathbf{y}_2) \right]. \quad (3.2)$$

Minimizing this term applies a repulsive force between the feature vectors of the chosen and rejected responses. Geometrically, this acts as a soft repulsion constraint, preventing the latent representations of distinct responses from collapsing into a single point, thereby maintaining the necessary volume in the embedding space for effective learning.

Final Objective. The dual-space S-SPPO objective at the

t -th iteration is a weighted sum of (3.1) and (3.2):

$$\mathcal{L}_{S\text{-SPPO}}(\pi_\theta; \pi_t) = \mathbb{E}_{\mathcal{D}_t} \left[\left(r_\theta(\mathbf{y}) - \frac{1}{\beta} \left(\widehat{\mathbb{P}}_c(\mathbf{y} \succ \pi_t | \mathbf{x}) - \frac{1}{2} \right) \right)^2 + \lambda \cdot h_\theta(\mathbf{y}_1)^\top h_\theta(\mathbf{y}_2) \right].$$

where λ is a weight balance parameter.

Rationale: Decoupling Calibration Sources. Crucially, our framework employs distinct metric sources to match the objective of each space. For Supervision Calibration, we utilize an *external* embedding model (Wang et al., 2020; Günther et al., 2023; Nussbaum et al., 2024) to establish a model-agnostic anchor, ensuring that semantic targets remain objective and stable throughout training. Conversely, Representation Calibration operates directly on the *policy’s* own hidden states. Since manifold collapse is an intrinsic degeneration of the feature space, we explicitly intervene within the policy’s latent geometry to enforce diversity where it matters most.

3.3. Theoretical Analysis

As discussed in Section 2, standard SPPO formulates preference optimization as a symmetric two-player constant-sum game at the *policy level*, where the payoff is defined by the expected win rate $\mathbb{P}(\pi \succ \pi' | \mathbf{x})$. Its convergence guarantees rely critically on the constant-sum structure (i.e., antisymmetry) of this game. S-SPPO modifies the empirical preference signal through semantic calibration. A natural theoretical concern is whether this calibration breaks the constant-sum property, potentially destabilizing the Nash Equilibrium. In this section, we rigorously prove that our semantic calibration mechanism preserves this essential game-theoretic structure, thereby suggesting that S-SPPO inherits the convergence guarantees of standard SPPO.

3.3.1. POLICY-LEVEL CONSTANT-SUM PRESERVATION

We analyze the properties of the calibrated preference probability $\widehat{\mathbb{P}}_c$ in the general pairwise setting.

Lemma 3.1 (Policy-Level Constant-Sum Preservation). *Consider the pairwise calibrated preference probability $\widehat{\mathbb{P}}_c$ consistent with the update rule in Eq. (3.1). For any pair of policies (π, π') and any prompt $\mathbf{x}$, the induced policy-level win rates satisfy:*

$$\mathbb{P}_c(\pi \succ \pi' | \mathbf{x}) + \mathbb{P}_c(\pi' \succ \pi | \mathbf{x}) = 1, \quad (3.3)$$

where $\mathbb{P}_c(\pi \succ \pi' | \mathbf{x}) \triangleq \mathbb{E}_{\mathbf{y} \sim \pi, \mathbf{y}' \sim \pi'} [\widehat{\mathbb{P}}_c(\mathbf{y} \succ \mathbf{y}' | \mathbf{x})]$. Consequently, semantic calibration induces a valid symmetric constant-sum game at the policy level.

Proof. See Appendix A.1. $\square$

Corollary 3.2 (Convergence of the Calibrated Game). *Let $\bar{\pi}_T = \frac{1}{T} \sum_{t=1}^T \pi_t$ denote the averaged policy produced by S-SPPO (considering the calibrated objective). Under standard realizability assumptions, the duality gap of $\bar{\pi}_T$ with respect to the Nash Equilibrium of the calibrated policy-level game decays at a rate of $O(1/\sqrt{T})$.*

Proof. See Appendix A.2. $\square$

Proposition 3.3 (ϵ -Nash Equilibrium of the Regularized Game). *Consider the regularized symmetric constant-sum game induced by S-SPPO, with the payoff function defined as:*

$$U_\lambda(\pi, \pi') = \mathbb{P}_c(\pi \succ \pi' | \mathbf{x}) - \lambda \mathcal{R}_{rep}(\pi) + \lambda \mathcal{R}_{rep}(\pi') \quad (3.4)$$

where $\mathbb{P}_c$ is the calibrated policy-level win rate, and $\lambda \mathcal{R}_{rep}$ denotes the geometric regularization term. Assuming the normalized latent embeddings bound the representation term such that $|\mathcal{R}_{rep}(\cdot)| \leq M$, let π_λ^* be the Nash Equilibrium of this regularized game. Then, π_λ^* constitutes an ϵ -Nash Equilibrium of the original calibrated game without latent repulsion, where the suboptimality gap is bounded by $\epsilon \leq 2\lambda M$.

Proof. See Appendix A.3. $\square$

4. Experiments

4.1. Experimental Setup

Models and Datasets. Following Wu et al. (2025), we conduct our experiments using two base models: Mistral-7B-Instruct-v0.2 (Jiang et al., 2023a) and Llama-3-8B-Instruct. For the dataset, we utilize UltraFeedback (Cui et al.) exclusively as a prompt source, extracting approximately 60k prompts while *discarding the accompanying responses*. Adopting the iterative strategy from SPPO (Wu et al., 2025), we split these prompts into three portions, using one portion per iteration to prevent overfitting. Following (Wu et al., 2025), we employ PairRM (Jiang et al., 2023b), a 0.4B parameter pairwise preference model, as our preference oracle to annotate generated responses and estimate win rates.

Baselines. We compare our method against a comprehensive set of baselines covering both base models and state-of-the-art alignment algorithms following Wu et al. (2025). First, we evaluate the instruction-tuned Base Models (Mistral-7B-Instruct-v0.2 and Llama-3-8B-Instruct) directly. Second, we benchmark against SPPO (Wu et al., 2025), which serves as our primary baseline. Third, for the sake of completeness, we implement Iterative DPO (Rafailov et al., 2023) and Iterative IPO (Azar et al., 2024) by aligning their response generation and training pipelines exactly with SPPO, differing solely in the substitution of the optimization

Table 1. AlpacaEval 2.0 evaluation against GPT-4 Turbo. We report results using both the GPT-4 Annotator and the GPT-4 Turbo Annotator (see Appendix B.4) as judges. For the SPPO baseline, we directly report the results from Wu et al. (2025) where available (i.e., GPT-4 Annotator); otherwise, we evaluate using the official released checkpoints (i.e., GPT-4 Turbo Annotator, see Appendix B.3). Across both evaluation protocols, S-SPPO consistently outperforms baselines in both raw and length-controlled win rates.

Model	GPT-4 Annotator			GPT-4 Turbo Annotator		
	LC Win	Win Rate	Avg. Len	LC Win	Win Rate	Avg. Len
Mistral-7B-Instruct-v0.2	17.11	14.72	1676	18.84	15.38	1622
Snorkel (Mistral-PairRM-DPO)	26.39	30.22	2736	-	-	-
Self-Rewarding 70B Iter1	-	9.94	1092	-	-	-
Self-Rewarding 70B Iter2	-	15.38	1552	-	-	-
Self-Rewarding 70B Iter3	-	20.44	2552	-	-	-
Mistral-7B-DPO Iter1	23.81	20.44	1723	-	-	-
Mistral-7B-DPO Iter2	24.23	24.46	2028	-	-	-
Mistral-7B-DPO Iter3	22.30	23.39	2189	-	-	-
Mistral-7B-IPO Iter1	23.78	20.77	1693	-	-	-
Mistral-7B-IPO Iter2	21.08	23.38	2660	-	-	-
Mistral-7B-IPO Iter3	20.06	22.47	2760	-	-	-
Mistral-7B-SPPO Iter1	24.79	23.51	1855	28.80	29.51	2028
Mistral-7B-SPPO Iter2	26.89	27.62	2019	32.60	35.31	2172
Mistral-7B-SPPO Iter3	28.53	31.02	2163	34.90	39.46	2294
Mistral-7B-S-SPPO Iter1	24.28	25.02	2039	28.34	29.18	2039
Mistral-7B-S-SPPO Iter2	28.22	31.42	2206	34.89	38.04	2206
Mistral-7B-S-SPPO Iter3	30.84	35.09	2326	36.13	40.26	2326
Llama-3-8B-Instruct	22.92	22.57	1899	28.31	28.52	1975
Llama-3-8B-SPPO Iter1	31.73	31.74	1962	35.69	36.29	2010
Llama-3-8B-SPPO Iter2	35.15	35.98	2021	42.24	43.42	2070
Llama-3-8B-SPPO Iter3	38.77	39.85	2066	44.79	46.84	2128
Llama-3-8B-S-SPPO Iter1	31.33	34.72	2114	38.18	40.37	2114
Llama-3-8B-S-SPPO Iter2	37.07	42.07	2208	44.77	48.47	2208
Llama-3-8B-S-SPPO Iter3	40.95	46.52	2287	47.46	52.19	2287

objective (i.e., replacing the SPPO loss with DPO or IPO loss). We also maintain the same model selection scheme (optimizing based on PairRM win rates) to ensure consistency. Additionally, we include Snorkel (Mistral-PairRM-DPO)(see Appendix B.3), a widely-used model trained via three rounds of iterative DPO using PairRM. Finally, we include Self-Rewarding LM (Yuan et al., 2024) as a representative of self-play methods that utilize the LLM itself as a judge.

Evaluation Benchmarks. We assess performance across four widely recognized benchmarks to evaluate diverse capabilities. To measure instruction-following proficiency in realistic user interaction scenarios, we report win rates and length-controlled win rates on AlpacaEval 2.0 (Dubois et al., 2024). We evaluate multi-turn reasoning and conversation quality on the rigorous MT-Bench (Zheng et al., 2023), and the judge directly assigns the score. For the SPPO baseline, we directly report the results from Wu et al. (2025) where available. For benchmarks not covered in the original paper,

we conduct evaluations using their official released checkpoints. Furthermore, for challenging open-ended queries, we utilize Arena-Hard (Li et al., 2025) (v0.1, consistent with SPPO setup), a benchmark distinguished by its high separability and strong correlation with human rankings in Chatbot Arena. We report with win rates.

Implementation Details. During the generation phase, we sample $K = 5$ responses per prompt strictly following the setup of SPPO (Wu et al., 2025). The winner and loser pairs for optimization are selected based on PairRM scores. Due to space constraints, detailed hyperparameters, probability estimation formulas, and extensive baseline configurations are provided in Appendix B.

4.2. Experimental Results

We evaluate S-SPPO across three standard benchmarks. In general, we observe that post-training alignment methods yield improvements over instruction-tuned base models.

Table 2. Comparisons on the AlpacaEval 2.0 Leaderboard. We compare our method against proprietary models (e.g., GPT-4 Turbo, Claude 3 Opus) and other top-tier open-source models.

Model	AlpacaEval 2.0	
	LC Win Rate	Win Rate
GPT-4 Turbo	50.0	50.0
Llama-3-8B-S-SPPO Iter3	41.0	46.5
Llama-3-8B-SPPO Iter3	38.8	39.9
Claude 3 Opus	40.5	29.1
GPT-4 0314	35.3	22.1
Llama 3 70B Instruct	34.4	33.2
Mistral-7B-S-SPPO Iter3	30.8	35.1
Mistral-7B-SPPO Iter3	28.5	31.0
GPT-4 0613	30.2	15.8
Mistral Medium	28.6	21.9
Claude 2	28.2	17.2
Snorkel (Mistral-PairRM-DPO)	26.4	30.2
Gemini Pro	24.4	18.2
Mistral 8×7B v0.1	23.7	18.1
Llama 3 8B Instruct	22.9	22.6

However, *S*-SPPO consistently achieves the most substantial gains among all compared methods, effectively addressing the stability issues inherent in standard self-play.

Instruction Following. As detailed in Table 1 and Table 2, while most iterative alignment baselines improve upon the base models, *S*-SPPO secures the largest performance leap. On Mistral-7B, for instance, our method effectively doubles the length-controlled (LC) win rate of the base model (from 14.72% to **30.84%**), outperforming the best result achieved by Snorkel (26.39%) and other baselines. When compared directly to standard SPPO, our method demonstrates consistent superiority across all iterations. On Llama-3-8B, *S*-SPPO reaches a final LC win rate of **40.95%**, surpassing SPPO’s 38.77%. This advantage is further validated on the AlpacaEval 2.0 Leaderboard, where Llama-3-8B-S-SPPO (41.0%) not only exceeds the SPPO baseline but also outperforms much larger proprietary models (e.g., Claude 3 Opus and GPT-4 0314). Most notably, when evaluated with the **GPT-4 Turbo** annotator, Llama-3-8B-S-SPPO achieves remarkable performance, reaching **52.19%** and **47.46%** in raw and LC win rates, respectively, setting a new standard for open-source models of this scale among self-play alignment methods without access to ground-truth references from the training dataset.

Multi-Turn Conversation. In multi-turn evaluations (Table 3), while some baselines struggle to maintain the conversational capabilities of the base model, *S*-SPPO demonstrates the highest ceiling. Specifically, on Llama-3-8B, *S*-SPPO achieves the highest average score of **8.22**. A critical comparison reveals the limitations of the uncalibrated baseline: standard SPPO exhibits a downward trend on Llama-3 (8.01 → 7.93) and an initial regression on Mistral-7B (falling below the base model to 7.21). In contrast,

Table 3. MT-Bench evaluation for multi-turn conversation capabilities. We focus on the average score. Unlike SPPO, which shows a downward trend across iterations on Llama-3 (8.01 → 7.93), *S*-SPPO demonstrates stable or increasing performance trends. Notably, *S*-SPPO achieves the highest average score, surpassing even the best-performing iteration of all baselines.

Model	MT-Bench		
	1st Turn	2nd Turn	Average
Mistral-7B-Instruct-v0.2	7.78	7.25	7.51
Snorkel (Mistral-PairRM-DPO)	7.83	7.33	7.58
Mistral-7B-DPO Iter1	7.45	6.58	7.02
Mistral-7B-DPO Iter2	7.57	6.56	7.06
Mistral-7B-DPO Iter3	7.49	6.69	7.09
Mistral-7B-SPPO Iter1	7.63	6.79	7.21
Mistral-7B-SPPO Iter2	7.90	7.08	7.49
Mistral-7B-SPPO Iter3	7.84	7.34	7.59
Mistral-7B-S-SPPO Iter1	7.84	7.26	7.55
Mistral-7B-S-SPPO Iter2	8.04	7.05	7.55
Mistral-7B-S-SPPO Iter3	7.96	7.36	7.66
Llama-3-8B-SPPO Iter1	8.29	7.73	8.01
Llama-3-8B-SPPO Iter2	8.33	7.61	7.97
Llama-3-8B-SPPO Iter3	8.36	7.49	7.93
Llama-3-8B-S-SPPO Iter1	8.20	7.85	8.03
Llama-3-8B-S-SPPO Iter2	8.60	7.82	8.21
Llama-3-8B-S-SPPO Iter3	8.54	7.90	8.22

S-SPPO effectively mitigates this degradation, yielding consistent performance gains across iterations. This suggests that our dual-space calibration successfully stabilizes the self-play optimization, preventing the rapid collapse often observed in the uncalibrated baseline.

Complex Open-Ended Queries. Finally, on the challenging Arena-Hard-Auto v0.1 benchmark (Table 4), we observe that while alignment generally improves performance over the base model (e.g., Mistral-7B improves from 12.6 to 23.9 with *S*-SPPO), *S*-SPPO again achieves the highest empirical results. Most notably, our method proves significantly more stable than standard SPPO. On Llama-3, standard SPPO suffers from a severe performance collapse, dropping from 31.0 to 29.8 across iterations. *S*-SPPO successfully avoids this overfitting, maintaining stable learning dynamics and outperforming the standard SPPO baseline on Mistral-7B (23.9 vs. 23.3), thereby confirming the necessity of dual-space calibration for handling high-complexity prompts.

4.3. Analysis

Impact of Dual-Space Calibration. To validate the individual and synergistic contributions of our proposed Dual-Space Semantic Calibration, we conduct an ablation study on Llama-3-8B by selectively removing the Supervision Calibration and Representation Calibration modules. As summarized in Table 5, the full *S*-SPPO framework achieves the highest performance with a peak length-controlled (LC)

Table 4. Arena-Hard-Auto evaluation. We observe that standard SPPO suffers from performance degradation on Llama-3 across iterations. In contrast, S-SPPO successfully mitigates this collapse, demonstrating consistent performance improvements.

Model	Arena-Hard-Auto
Mistral-7B-Instruct	12.6
Snorkel (Mistral-PairRM-DPO)	20.7
Mistral-7B-SPPO Iter1	18.7
Mistral-7B-SPPO Iter2	20.4
Mistral-7B-SPPO Iter3	23.3
Mistral-7B-S-SPPO Iter1	21.5
Mistral-7B-S-SPPO Iter2	21.8
Mistral-7B-S-SPPO Iter3	23.9
Llama-3-8B-SPPO Iter1	31.0
Llama-3-8B-SPPO Iter2	30.1
Llama-3-8B-SPPO Iter3	29.8
Llama-3-8B-S-SPPO Iter1	30.0
Llama-3-8B-S-SPPO Iter2	30.6
Llama-3-8B-S-SPPO Iter3	31.5

Table 5. Ablation study on Dual-Space Semantic Calibration. We analyze the impact of removing Supervision Calibration and Representation Calibration. The results demonstrate that both components are essential, with the full S-SPPO framework achieving the highest win rates.

Supervision Calibration	Representation Calibration	LC Win Rate		
		Iter. 1	Iter. 2	Iter. 3
✗	✗	35.69	42.24	44.79
✓	✗	38.39	43.91	46.16
✗	✓	37.01	43.79	45.20
✓	✓	38.18	44.77	47.46

win rate of **47.46%**. In contrast, the uncalibrated baseline (Row 1) lags significantly behind at 44.79%. Introducing components individually reveals that both are indispensable: applying Supervision Calibration alone (Row 2) improves performance to 46.16% by filtering noisy preference labels, while Representation Calibration alone (Row 3) yields 45.20% by preserving latent diversity between chosen and rejected samples. These results confirm that S-SPPO’s superior performance stems from the joint effect of calibrating optimization targets and regularizing the feature space, rather than any single mechanism in isolation.

Additional Analysis. We further investigate the sensitivity of the encoder $\phi(\cdot)$ in Appendix C.1, computational efficiency in Appendix C.2, and the sensitivity of λ in Appendix C.3.

5. Related Work

Reinforcement Learning from Human Feedback (RLHF, Christiano et al., 2017; Ouyang et al., 2022) has emerged as a promising paradigm for aligning LLMs with human preference. Common RLHF approaches, such as Proximal

Policy Optimization (PPO, Schulman et al., 2017), typically rely on an explicit reward model, which can be difficult to design and reliably validate in many settings. DPO (Rafailov et al., 2023) mitigates this challenge by using LLMs as implicit rewards. We, henceforth, direct our main focus on this family of methods.

Preference Optimization Objectives. Many efforts have been devoted to different preference optimization objectives beyond DPO. SLic-HF (Zhao et al., 2023) introduces a margin-calibrated Slic loss that enforces a winner–loser separation. IPO (Azar et al., 2024) proposes a KL-regularized preference-optimization objective that applies an identity transform to fit pairwise preferences directly, without relying on the standard Bradley–Terry formulation. KTO (Ethayarajh et al., 2024) derives a preference-optimization loss from Kahneman–Tversky utility, increasing scores for “kept” responses and decreasing them for “rejected” ones. Odds Ratio Preference Optimization (ORPO, Hong et al., 2024) enables supervised fine-tuning and preference alignment in a single training run, without requiring an intermediate reference policy. Simple Preference Optimization (SimPO, Meng et al., 2024) introduces a reference-free objective that mitigates length bias by maximizing the gap between the preferred and dispreferred responses’ average log-likelihood.

Offline and Online Preference Optimization. Most prior DPO variants operate in an offline setting (Rafailov et al., 2023; Ethayarajh et al., 2024; Hong et al., 2024; Meng et al., 2024), optimizing solely on a fixed and pre-collected preference dataset. However, without an explicit reward model, they cannot sample preference pairs from the optimal policy, which hinders their ability to capture the intransitivity and irrationality in human preferences. To address this, early work augments the preference dataset by sampling from a trained SFT policy (Zhao et al., 2023) or from a further-refined SFT policy via rejection sampling (Liu et al., 2024), enabling learning from data that more closely reflects the optimal policy’s behavior. Recently, many studies have extended this idea to an iterative, online setting, either by continually updating the reference model to the latest policy or by generating fresh preference pairs at each training iteration (Xiong et al., 2024; Dong et al., 2024; Rosset et al., 2024; Kim et al., 2025; Wu et al., 2025). Concurrently, various works have explored alternative game-theoretic formulations (Wang et al., 2025; Pásztor et al., 2025; Tang et al., 2025) or autonomous self-improvement mechanisms (Ji et al., 2024; Chen et al., 2024; Le et al., 2025; Xiang et al., 2025) to enhance stability. In this paper, we focus specifically on Self-Play Preference Optimization (SPPO, Wu et al., 2025) because it is one of the most fundamental works for addressing intransitivity and irrationality by directly optimizing preference probabilities within a constant-sum game framework, rather than relying on reward maximization. Moreover, it generates on-policy self-play training signals using a lightweight preference

model, without requiring human-annotated preference data.

6. Conclusion

In this work, we identify a critical instability in self-play preference optimization: policy degeneration occurs when preference oracles assign overconfident wins to semantically indistinguishable responses. To address this, we propose S-SPPO, a dual-space calibration framework that stabilizes alignment via Supervision Calibration (semantic gating) and Representation Calibration (latent repulsion). Theoretically, we prove that S-SPPO preserves the underlying constant-sum game structure, theoretically supporting convergence to a Nash Equilibrium. Empirically, S-SPPO sets a new standard for open-source models, achieving a 52.19% win rate and 47.46% length-controlled win rate on AlpacaEval 2.0 without relying on human-annotated data. We provide further discussion on limitations and future directions in Appendix D.

Impact Statement

This paper addresses the instability in Large Language Model alignment. We see three main potential impacts of our work:

Improved Reliability. Standard SPPO paradigm often forces models to fit noise in the reward model. By neutralizing supervision signals for semantically indistinguishable pairs, our method discourages the policy from overfitting to spurious preferences. This leads to models that are less likely to hallucinate differences in ambiguous situations.

Preserved Diversity. Alignment training frequently causes *mode collapse*, where models become repetitive. Our method forces the model to maintain distinct internal representations. This ensures the model retains its variety and does not degenerate into a narrow set of outputs.

Reduced Reliance on Human Data. Self-play optimization generates training signals automatically using a reward model. This removes the need for expensive human-annotated preference labels. By making this process stable and effective, our framework allows models to improve autonomously without the high cost of manual data collection.

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A. Proof

A.1. Proof of Lemma 3.1

Proof. By definition, the policy-level win rate is the expectation of the calibrated pairwise preference. Thus, it suffices to show that the calibrated pairwise probability $\hat{\mathbb{P}}_c$ maintains the antisymmetry property of the original preference oracle.

For any response pair $(\mathbf{y}_i, \mathbf{y}_j)$, let $S_{ij} = S_{ji}$ denote their semantic similarity. The calibrated pairwise probability is given by:

$$\hat{\mathbb{P}}_c(\mathbf{y}_i \succ \mathbf{y}_j) = \frac{1}{2} + (1 - S_{ij}) \left(\hat{\mathbb{P}}(\mathbf{y}_i \succ \mathbf{y}_j) - \frac{1}{2} \right). \quad (\text{A.1})$$

We examine the reverse pair $(\mathbf{y}_j, \mathbf{y}_i)$. Since the original preference oracle is a valid probability distribution, it satisfies $\hat{\mathbb{P}}(\mathbf{y}_j \succ \mathbf{y}_i) = 1 - \hat{\mathbb{P}}(\mathbf{y}_i \succ \mathbf{y}_j)$. Substituting this into the calibration formula:

$$\begin{aligned} \hat{\mathbb{P}}_c(\mathbf{y}_j \succ \mathbf{y}_i) &= \frac{1}{2} + (1 - S_{ji}) \left(\hat{\mathbb{P}}(\mathbf{y}_j \succ \mathbf{y}_i) - \frac{1}{2} \right) \\ &= \frac{1}{2} + (1 - S_{ij}) \left(\left(1 - \hat{\mathbb{P}}(\mathbf{y}_i \succ \mathbf{y}_j) \right) - \frac{1}{2} \right) \\ &= \frac{1}{2} + (1 - S_{ij}) \left(\frac{1}{2} - \hat{\mathbb{P}}(\mathbf{y}_i \succ \mathbf{y}_j) \right) \\ &= \frac{1}{2} - (1 - S_{ij}) \left(\hat{\mathbb{P}}(\mathbf{y}_i \succ \mathbf{y}_j) - \frac{1}{2} \right) \\ &= 1 - \left[\frac{1}{2} + (1 - S_{ij}) \left(\hat{\mathbb{P}}(\mathbf{y}_i \succ \mathbf{y}_j) - \frac{1}{2} \right) \right] \\ &= 1 - \hat{\mathbb{P}}_c(\mathbf{y}_i \succ \mathbf{y}_j). \end{aligned}$$

Since $\hat{\mathbb{P}}_c(\mathbf{y}_i \succ \mathbf{y}_j) + \hat{\mathbb{P}}_c(\mathbf{y}_j \succ \mathbf{y}_i) = 1$ holds for any realization $(\mathbf{y}_i, \mathbf{y}_j)$, taking the expectation over $\mathbf{y} \sim \pi(\cdot|\mathbf{x})$ and $\mathbf{y}' \sim \pi'(\cdot|\mathbf{x})$ preserves this equality by linearity. Thus, the policy-level game remains constant-sum. $\square$

A.2. Proof of Corollary 3.2

Proof. Lemma 3.1 establishes that the semantic calibration preserves the symmetric constant-sum structure. Accordingly, the S-SPPO update can be interpreted as approximating multiplicative weight updates (MWU) on this calibrated game. Standard no-regret learning results for constant-sum games (Freund & Schapire, 1999; Wu et al., 2025) apply directly, facilitating convergence to the Nash Equilibrium of the calibrated game defined by $\mathbb{P}_c$. $\square$

A.3. Proof of Proposition 3.3

Proof. The asymmetric construction of the regularization terms ensures that $U_\lambda(\pi, \pi') + U_\lambda(\pi', \pi) = 1$, preserving the strictly constant-sum property. By the definition of the Nash Equilibrium for U_λ , it holds that $U_\lambda(\pi, \pi_\lambda^*) \leq 1/2$ for any policy π . Substituting the payoff definition yields:

$$\mathbb{P}_c(\pi > \pi_\lambda^* | x) - \lambda \mathcal{R}_{rep}(\pi) + \lambda \mathcal{R}_{rep}(\pi_\lambda^*) \leq \frac{1}{2} \quad (\text{A.2})$$

Rearranging the terms, we obtain:

$$\mathbb{P}_c(\pi > \pi_\lambda^* | x) \leq \frac{1}{2} + \lambda (\mathcal{R}_{rep}(\pi) - \mathcal{R}_{rep}(\pi_\lambda^*)) \quad (\text{A.3})$$

Given the boundedness assumption $|\mathcal{R}_{rep}(\cdot)| \leq M$, the maximum deviation from the original equilibrium (1/2) is strictly bounded by:

$$\mathbb{P}_c(\pi > \pi_\lambda^* | x) \leq \frac{1}{2} + \lambda (M - (-M)) = \frac{1}{2} + 2\lambda M \quad (\text{A.4})$$

This concludes the proof, demonstrating that the theoretical deviation from the unregularized Nash Equilibrium is linearly bounded by the regularization weight λ . $\square$

Algorithm 1 Algorithmic Summary of proposed S-SPPO.

- 1: **input:** base policy π_{θ_1} , preference oracle $\mathbb{P}$, semantic encoder $\phi(\cdot)$, scaling factor $\frac{1}{\beta}$, regularization weight λ , number of generated samples K .
- 2: **for** $t = 1, 2, \dots$ **do**
- 3: Generate responses $\mathbf{y}_{1:K} \sim \pi_t(\cdot|\mathbf{x})$ for prompt $\mathbf{x} \sim \mathcal{X}$.
- 4: Annotate win rates $\hat{P}(\mathbf{y} \succ \pi_t|\mathbf{x})$ for all $\mathbf{y} \in \mathbf{y}_{1:K}$.
- 5: **Selection:** Identify winner $\mathbf{y}_w$ and loser $\mathbf{y}_l$ based on scores.
 Supervision Calibration: Compute similarity $S = \text{sim}(\phi(\mathbf{y}_w), \phi(\mathbf{y}_l))$ and calibrate targets for both $y \in \{\mathbf{y}_w, \mathbf{y}_l\}$:

$$\hat{P}_c(\mathbf{y}) \leftarrow \frac{1}{2} + (1 - S) \left(\hat{P}(\mathbf{y}) - \frac{1}{2} \right)$$

(Symmetrically pulls both winner and loser targets towards 0.5)
- 6: Form dataset $\mathcal{D}_t = \{(\mathbf{x}, \mathbf{y}, \hat{P}_c(\mathbf{y})) \mid \mathbf{y} \in \{\mathbf{y}_w, \mathbf{y}_l\}\}$.
- 7: Optimize $\pi_{\theta_{t+1}}$ with **Representation Calibration:**

$$\theta_{t+1} \leftarrow \underset{\theta}{\operatorname{argmin}} \mathbb{E}_{(\mathbf{x}, \mathbf{y}, \hat{P}_c) \sim \mathcal{D}_t} \left[\underbrace{\left(\log \frac{\pi_{\theta}(\mathbf{y}|\mathbf{x})}{\pi_t(\mathbf{y}|\mathbf{x})} - \frac{1}{\beta} \left(\hat{P}_c - \frac{1}{2} \right) \right)^2}_{\text{Regression on Calibrated Targets}} + \lambda \cdot h_{\theta}(\mathbf{y}_w)^{\top} h_{\theta}(\mathbf{y}_l) \right]$$
- 9: **end for**

B. Detailed Experimental Settings

B.1. Training Hyperparameters

All experiments are conducted on $8 \times$ Nvidia H100 GPUs. For our proposed method (S-SPPO), we train for a total of three iterations. In each iteration, we select the checkpoint with the highest average win rate (judged by PairRM-0.4B on a hold-out subset of UltraFeedback) to proceed to the next round. We trained the model for 18 epochs using a global batch size of 64 and we set $\beta = 1\text{e} - 3$, with a learning rate of $5.0\text{e} - 7$ that follows a linear decay schedule with a warmup ratio of 0.1. These settings were selected to ensure stability and fair comparison across all iterative baselines (S-SPPO, Iterative DPO, Iterative IPO). λ is set to 1. We choose `all-MiniLM-L6-v2` as our default lightweight encoder model.

B.2. Response Generation and Probability Estimation

During the generation phase of each iteration, we sample $K = 5$ responses for each prompt from the current policy using a temperature of 1.0 and top- $p = 1.0$. To estimate the winning probability distribution $\hat{P}(\mathbf{y}_i \succ \pi_t|\mathbf{x})$, we calculate the average win rate of a response $\mathbf{y}_i$ against all other sampled responses $\mathbf{y}_k$:

$$\hat{P}(\mathbf{y}_i \succ \pi_t|\mathbf{x}) = \frac{1}{K} \sum_{k=1}^K \mathbb{P}(\mathbf{y}_i \succ \mathbf{y}_k|\mathbf{x}), \quad \forall i \in [K]. \quad (\text{B.1})$$

The pairwise probability $\mathbb{P}(\mathbf{y} \succ \mathbf{y}'|\mathbf{x})$ is derived from the PairRM scoring function $s(\mathbf{y}, \mathbf{y}'; \mathbf{x})$, which reflects the relative strength difference:

$$\mathbb{P}(\mathbf{y} \succ \mathbf{y}'|\mathbf{x}) = \frac{\exp(s(\mathbf{y}, \mathbf{y}'; \mathbf{x}))}{1 + \exp(s(\mathbf{y}, \mathbf{y}'; \mathbf{x}))}. \quad (\text{B.2})$$

Unlike Bradley-Terry models which assume transitivity, PairRM assigns relative rewards directly, allowing it to model general, potentially non-transitive preferences. For preference pair construction, following Snorkel (see next section for the link), we select the response with the highest PairRM score as the winner $\mathbf{y}_w$ and the lowest as the loser $\mathbf{y}_l$, where the score

is defined as:

$$s_{\text{PairRM}}(\mathbf{y}_i; \mathbf{x}) := \frac{1}{K} \sum_{k=1}^K s(\mathbf{y}_i, \mathbf{y}_k; \mathbf{x}). \quad (\text{B.3})$$

B.3. Baseline Model and Encoder Details

- Mistral-7B-Instruct-v0.2: <https://huggingface.co/mistralai/Mistral-7B-Instruct-v0.2>.
- Llama-3-8B-Instruct: <https://huggingface.co/meta-llama/Meta-Llama-3-8B-Instruct>.
- Snorkel (Mistral-PairRM-DPO): <https://huggingface.co/snorkelai/Snorkel-Mistral-PairRM-DPO>.
- PairRM: <https://huggingface.co/llm-blender/PairRM>.
- SPPO official checkpoints: <https://huggingface.co/collections/UCLA-AGI/sppo>.
- all-MiniLM-L6-v2: <https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>. It has a model size of 22.7M.
- jina-embeddings-v3 (Sturua et al., 2024): <https://huggingface.co/jinaai/jina-embeddings-v3>. It has a model size of 0.6B.
- nomic-ai/nomic-embed-text-v1.5 (Nussbaum et al., 2024): <https://huggingface.co/nomic-ai/nomic-embed-text-v1.5>. It has a model size of 0.1B.

B.4. Benchmark Details

- **AlpacaEval 2.0:** Uses AlpacaFarm prompts (Dubois et al., 2023). Responses are judged against a GPT-4-Turbo baseline using different annotators, i.e, judges. We report the length-controlled win rate. We provided their link here:
 - GPT-4 annotator: https://github.com/tatsu-lab/alpaca_eval/blob/main/src/alpaca_eval/evaluators_configs/weighted_alpaca_eval_gpt4_turbo/configs.yaml.
 - GPT-4 Turbo annotator: https://github.com/tatsu-lab/alpaca_eval/blob/main/src/alpaca_eval/evaluators_configs/weighted_alpaca_eval_gpt4_turbo_new/configs.yaml.
- **Arena-Hard:** A dataset designed to have high separability and correlation with Chatbot Arena Elo ratings. It focuses on complex, open-ended queries (Li et al., 2025).
- **MT-Bench:** Consists of 80 high-quality multi-turn questions covering coding, reasoning, and role-playing. Responses are graded on a scale of 1-10 by GPT-4 (Zheng et al., 2023).

C. More Analysis

C.1. Sensitivity of Encoder $\phi(\cdot)$

In Fig. 2, we use the encoder all-MiniLM-L6-v2 to reflect the uncalibrated nature of the original SPPO paper (Wu et al., 2025). Here, we demonstrate that this observation is consistent across different encoder architectures. To illustrate this, we plot the features extracted by representative encoders in Fig. 3, which reveals the same pattern of misalignment: the preference oracle assigns high confidence even when semantic similarity is high. Furthermore, as shown in Table 6, we observe that while the choice of encoder leads to minor fluctuations in final performance, our proposed S-SPPO consistently outperforms the baseline across all iterations.

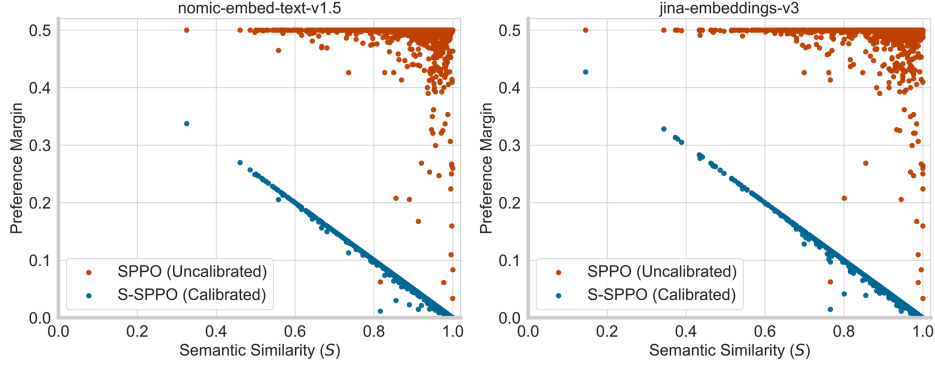


Figure 3. Uncalibration across different encoders. We visualize the preference margin against semantic similarity for `nomic-embed-text-v1.5` and `jina-embeddings-v3`. Similar to the observations with `all-MiniLM-L6-v2`, the uncalibrated baseline (SPPO) fails to converge to a neutral preference margin even for semantically indistinguishable pairs ($S \approx 1.0$), whereas S-SPPO successfully calibrates the margin.

Table 6. Ablation study on encoder selection. We report the Length-Controlled (LC) win rates on AlpacaEval 2.0. S-SPPO achieves consistent gains over the baseline regardless of the specific semantic encoder used for calibration.

	Iter. 1	Iter. 2	Iter. 3
Baseline	35.7	42.2	44.8
jina-embeddings-v3	36.8	44.1	47.7
nomic-embed-text-v1.5	39.0	43.9	46.9
all-MiniLM-L6-v2	38.2	44.8	47.5

C.2. Computational Efficiency Analysis

We assess the computational overhead of S-SPPO compared to the SPPO baseline. The implementation pipeline follows the procedure outlined in Algorithm 1, comprising three main phases: *Generation & Annotation* (Steps 3–4), *Preference Construction* (Steps 5–7), and *Policy Optimization* (Step 8). As shown in Table 7, the overhead introduced by our dual-space calibration is minor. During the *Preference Construction* phase, applying Supervision Calibration with a lightweight encoder adds only ~ 5 seconds to the runtime (90s $\rightarrow$ 95s) with a minimal GPU memory footprint of ~ 1.7 GB. In the *Policy Optimization* phase, Representation Calibration requires accessing the model’s hidden states, resulting in a *slight increase* in peak GPU memory (67.0GB $\rightarrow$ 72.8GB) due to computation graph storage, but virtually no increase in training time (~ 16 mins). Crucially, the *Total Pipeline* runtime remains approximately 66 minutes, as it is dominated by the generation and annotation phases (Steps 3–4) which are shared between methods. This confirms that S-SPPO achieves significant performance gains with minimal impact on training efficiency.

Table 7. Computational efficiency comparison between SPPO and S-SPPO on Llama-3-8B per iteration. The **Total** runtime includes the dominant *Generation & Annotation* phases (Steps 3–5 in Alg. 1), which are identical for both methods (~ 49 mins).

Phase	Preference Construction	Policy Optimization	Total Pipeline
Alg. Steps	(Steps 5–7)	(Step 8)	(Steps 3–7)
Runtime			
SPPO (Baseline)	~ 90 s	~ 16 min	~ 66 min
S-SPPO (Ours)	~ 95 s	~ 16 min	~ 66 min
Peak GPU Memory			
SPPO (Baseline)	~ 0 GB	67.0 GB	–
S-SPPO (Ours)	~ 1.7 GB	72.8 GB	–

C.3. Sensitivity of Hyperparameter λ

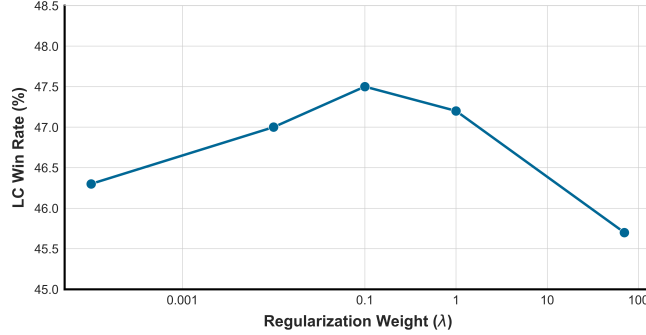


Figure 4. Sensitivity analysis of λ on AlpacaEval 2.0.

In this section, we analyze the sensitivity of *S*-SPPO to the regularization weight λ , which controls the strength of Representation Calibration. As illustrated in Fig. 4, we observe an inverted-U trend where performance peaks when λ is in the moderate range (around 0.1 to 1). When λ is too small, the repulsive force is insufficient, limiting the model’s discriminative power. Conversely, performance degrades as λ becomes excessively large. We attribute this to the disruption of the natural semantic manifold: since both chosen and rejected responses are generated by a capable policy, they naturally share semantic proximity. Forcing these responses to be strictly orthogonal via a large λ imposes an unnatural geometric constraint that hinders learning. Consequently, a moderate λ strikes the optimal balance, enforcing sufficient latent diversity without shattering the semantic structure of the policy.

D. Discussion

D.1. Theoretical Justification via Flat Reward Landscapes

Recent theoretical studies (Lee & Yoon, 2025) suggest that a robust reward function should exhibit a flat reward landscape in the policy parameter space to ensure robustness against action or reward perturbations. In the context of Large Language Model (LLM) preference optimization, when two generated responses are semantically near-identical (i.e., with extremely high semantic similarity S), they essentially represent minor perturbations in the action space. Consequently, an ideal and well-calibrated preference oracle should maintain a flat evaluation landscape across these local variations, naturally outputting a neutral preference probability close to 0.5. In practice, however, uncalibrated reward models severely violate this property, frequently assigning extreme win rates to semantically indistinguishable response pairs. This miscalibration injects sharp, high-magnitude reward gradients onto a landscape that should theoretically be flat, thereby destabilizing the self-play optimization trajectory. *S*-SPPO directly rectifies this fundamental flaw by progressively annealing the preference signal toward the maximum-entropy baseline (0.5) as semantic similarity increases. This strictly enforces the theoretically required flatness while preserving effective gradients for response pairs that exhibit genuine quality differences.

D.2. Safety Mechanism and Reasoning Domain Performance

A potential concern with incorporating general-purpose semantic encoders into preference optimization is their lack of fine-grained, token-level sensitivity in rigorous domains such as mathematical reasoning or coding. In these fields, a minor syntactic shift (e.g., a single negation sign) can completely alter the underlying utility. *S*-SPPO naturally guards against this risk through its structural safety design: the preference oracle exclusively dictates the optimization direction (i.e., which response wins), while the semantic calibration only conservatively scales the gradient magnitude toward 0.5. In the worst-case scenario, where the encoder mistakenly identifies two mathematically distinct responses as highly similar, *S*-SPPO merely dampens the training signal, leading to a more conservative optimization update rather than reversing the direction of optimization.

To empirically validate this safety property, we evaluate our model checkpoints on the GSM8K benchmark using strict-match accuracy. As shown in Table 8, *S*-SPPO not only preserves but slightly improves mathematical reasoning capabilities across different base models. This conclusively demonstrates that our semantic calibration does not induce vulnerabilities or

Table 8. GSM8K Benchmark Performance (Strict-match)

Model	GSM8K (Strict-match)
Mistral-7B Base	42.8%
Mistral-7B SPPO Iter3	44.0%
Mistral-7B S-SPPO Iter3	44.2%
Llama-3-8B Base	76.4%
Llama-3-8B SPPO Iter3	78.2%
Llama-3-8B S-SPPO Iter3	78.9%

performance degradation in reasoning tasks.

D.3. Scope and Domain-Specific Limitations

While our empirical results demonstrate that S-SPPO maintains safety on reasoning benchmarks, we explicitly scope our core contribution to general natural language preference alignment tasks. Applying this framework to highly rigorous domains, such as advanced mathematics or complex code generation, introduces distinct challenges that warrant careful consideration. Most off-the-shelf external embedding models are inherently optimized for general natural language semantics. Consequently, they often lack the fine-grained, token-level sensitivity required for these specialized tasks, where a minimal syntactic shift, such as a single negation sign or a modified conditional statement, can drastically alter the underlying semantic meaning and functional utility. Therefore, extending our framework to these specialized domains, potentially through the integration of domain-specific encoders or syntax-aware calibration metrics, remains an important avenue for future investigation.

D.4. The Necessity of Semantic Calibration with Advanced Reward Models

While state-of-the-art reward models can partially mitigate calibration issues, frameworks like S-SPPO remain fundamentally necessary for real-world alignment pipelines due to three primary considerations. First, regarding the inherent gap to a perfect oracle, current reward models still exhibit overconfidence and calibration issues when evaluating semantically similar responses. A theoretically flawless preference oracle is hypothetical; in practice, human preferences are intrinsically noisy, making explicit semantic calibration a necessary structural safeguard against persistent miscalibration. Second, S-SPPO inherently aligns with the premise of the SPPO paradigm. The fundamental motivation behind Self-Play Preference Optimization is that human preferences are profoundly challenging to model perfectly. If a flawless oracle existed, algorithmic distinctions between various advanced preference optimization methods would largely diminish. S-SPPO is designed specifically to address and correct these inevitable imperfections of practical reward models. Finally, computational efficiency in iterative self-play remains a critical bottleneck. Iterative self-play requires generating K candidates and performing $O(K^2)$ pairwise evaluations per prompt. Scaling this quadratic complexity with massive state-of-the-art models, such as 100B parameter models or LLM as a judge pipelines, over extensive datasets is computationally prohibitive. S-SPPO resolves this performance and computation tradeoff by enabling highly efficient and lightweight reward models, such as the 0.4B parameter PairRM, to achieve robust alignment, making extensive self-play scaling computationally feasible without astronomical inference costs.

D.5. Future Work

While S-SPPO demonstrates the effectiveness of dual-space semantic calibration, our exploration opens several avenues for future research. First, regarding the regularization weight λ , our current implementation maintains a fixed value across all iterations to demonstrate the robustness of the method without extensive hyperparameter tuning. However, it is plausible that a dynamic schedule of λ could further refine the equilibrium state (we observed some gain on this), though we prioritized simplicity and establishing a proof-of-concept over heavy tuning in this work. Second, we present a specific instantiation of calibration via linear semantic gating and Euclidean latent repulsion, yet this is merely one possibility within a broader design space. Future work could explore alternative calibration mechanisms, such as non-linear gating functions, uncertainty-aware calibration leveraging epistemic uncertainty, or kernel-based measures to capture more complex semantic dependencies. Finally, our current framework utilizes a fixed, off-the-shelf semantic encoder to guide the policy’s latent geometry. A

promising direction would be to more explicitly model the relationship between the external semantic space and the model’s internal representation, potentially by learning a projection that aligns the encoder’s embedding manifold directly with the policy’s latent space or by fine-tuning the encoder end-to-end to better capture the nuances of the specific task distribution.